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Preface

These notes cover the Linear Algebra course taught during Aug-Nov 2026 at CMI by Prof. Upendra. Throughout the course the emphasis will be on linear algebra over real and complex fields.

Timings

3 : 30 – 4 : 45pm Wednesday, Thursday, Friday. Shared tutorials with Analysis every Monday and Friday at 2pm.

Textbook

Algebra by Michael Artin, second edition. Other reference material: *Linear Algebra* by Kenneth Hoffman and Rau Kunze, *Linear Algebra Done Right* by Sheldon Axler.

Syllabus

Topics covered will be — Systems of linear equations, row reduction, vector spaces over any field, bases and dimension, change of basis, linear transformations, dimension formula, matrix of a linear transformation and change of bases, linear operators, similarity, determinants and invertibility, eigenvalues and eigenspaces, characteristic polynomial, triangular and diagonal forms, Euclidean/Hermitian spaces with standard inner product, orthogonal projection, orthonormal bases and Gram-Schmidt procedure, spectral theorem for Hermitian/symmetric operators. The above material corresponds to the following from Artin's Algebra: chapters 1, 3, 4.1–4.6, parts of chapter 8.

Grading

Homework will *not* be graded for the most part. He argues that this is due to LLMs, and this seems reasonable to me. Loosely, the grading will be based on two-three quizzes, two tests, and a final exam. I don't remember the exact distribution but it was something like 5 – 10% for homework, 10 – 20% for the quizzes, 30% for the tests and 30 – 40% for the final exam. Attendance will not be part of the grade, but is strongly encouraged.

Solving Linear Equations

1.1 Introduction

Our goal of this chapter is going to be solving linear equations. A linear equation in c variables can be written as

$$a_1x_1 + a_2x_2 + \cdots + a_cx_c = b,$$

where a_i and b are constants. There are multiple reasons that this is important, for one, in many non-linear cases, we can often get some understanding of the situations by looking at it locally, where it reduces to a linear system. One example is looking “near” the function in calculus, where it can be approximated to a linear function,

$$f(x) \approx f(0) + f'(0)x$$

near 0.

Regardless, our objective is going to be able to solve *systems of equations*, understand how to conceptualise the solutions, and be able to efficiently solve a given system of equations.

Consider the following system of equations— r linear equations in c variables:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1c}x_c &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2c}x_c &= b_2, \\ &\vdots \\ a_{r1}x_1 + a_{r2}x_2 + \cdots + a_{rc}x_c &= b_r. \end{aligned} \tag{1.1}$$

A solution to such a system is a c -tuple, i.e a member of the Cartesian product $\mathbb{R}^c$, which we shall see is the same thing as a column vector of length c .

How might we start solving such a system? Let us first consider an example and see how we reduce these equations.

Example 1.1

Consider the following two equations:

$$2x + 4y + 6z + 8w = 1 \quad (1)$$

$$3x + 0y + 7z + w = 10 \quad (2)$$

We can solve this by how we go about solving linear equations in highschool:
Dividing (1) by 2:

$$x + 2y + 3z + 4w = 1/2 \quad (3)$$

Subtracting 3 times (3) from (2):

$$\begin{aligned} (3x + 7z + w) - 3(x + 2y + 3z + 4w) &= 10 - 3 \cdot 1/2 \\ -6y - 2z - 11w &= 17/2 \end{aligned} \quad (4)$$

Dividing (4) by -6 :

$$y + 1/3 z + 11/6 w = -17/12 \quad (5)$$

Subtracting 2 times (5) from (3):

$$\begin{aligned} (x + 2y + 3z + 4w) - 2(y + 1/3 z + 11/6 w) &= 1/2 - 2(-17/12) \\ x + 7/3 z + 1/3 w &= 10/3 \end{aligned} \quad (6)$$

Finally, we get our solution by treating z and w as “free variables”, that is z, w ranged over any value in $\mathbb{R}^2$ and x, y are determined by z and w :

$$x = 10/3 - 7/3 z - 1/3 w$$

$$y = -17/12 - 1/3 z - 11/6 w$$

Note that at each step, we are changing our system of equations to an *equivalent system* which is easier to solve.

There is a certain set of operations that we can do to simplify our job, as we shall see in a minute. But first we will simplify our notation.

1.2 Matrices

Matrices

A $r \times c$ matrix, $A_{r \times c}$ is a rectangular array of r rows and c columns:

$$A_{r \times c} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1c} \\ a_{21} & a_{22} & \cdots & a_{2c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r1} & a_{r2} & \cdots & a_{rc} \end{pmatrix} \quad (1.2)$$

We denote the entry in the i -th row and j -th column as a_{ij} , in lower case, while we will denote the matrix itself in upper case.

Definition 1.1

We consider some operations that we can perform on matrices. In particular, we can add, scale, and multiply matrices. Addition and multiplication are defined component-wise, while scaling is defined by multiplying each entry by a scalar.

Addition of two matrices is defined for matrices of the same size, and is performed by adding the corresponding entries component-wise:

$$A_{r \times c} + B_{r \times c} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1c} + b_{1c} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2c} + b_{2c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r1} + b_{r1} & a_{r2} + b_{r2} & \cdots & a_{rc} + b_{rc} \end{pmatrix} \quad (1.3)$$

Scaling of a matrix is defined by multiplying each entry by a scalar¹:

$$\lambda A_{r \times c} = \begin{pmatrix} \lambda a_{11} & \lambda a_{12} & \cdots & \lambda a_{1c} \\ \lambda a_{21} & \lambda a_{22} & \cdots & \lambda a_{2c} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{r1} & \lambda a_{r2} & \cdots & \lambda a_{rc} \end{pmatrix} \quad (1.4)$$

Finally, matrix multiplication of two matrices $A_{r \times c}$ and $B_{c \times d}$ is defined by taking the dot product of each row of A with each column of B :

$$A_{r \times c} \cdot B_{c \times d} = \begin{pmatrix} \sum_{k=1}^c a_{1k} b_{k1} & \sum_{k=1}^c a_{1k} b_{k2} & \cdots & \sum_{k=1}^c a_{1k} b_{kd} \\ \sum_{k=1}^c a_{2k} b_{k1} & \sum_{k=1}^c a_{2k} b_{k2} & \cdots & \sum_{k=1}^c a_{2k} b_{kd} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{k=1}^c a_{rk} b_{k1} & \sum_{k=1}^c a_{rk} b_{k2} & \cdots & \sum_{k=1}^c a_{rk} b_{kd} \end{pmatrix} \quad (1.5)$$

Note that the number of columns in A must be equal to the number of rows in B .

Some properties of these operations are:

- Addition of matrices is commutative and associative.
- Multiplication of matrices is associative, but not commutative. In fact BA is not even defined in general, if AB is.

In reality it is often better to think of matrices really as “linear maps”, and this rectangular array as a *representation* of such a linear map. We shall go over this later.

1. A scalar for now is just a real or complex number. In general a scalar is any element of the field $\mathbb{F}$.

- Multiplication is distributive over addition, i.e. $A(B + C) = AB + AC$ for matrices A , B and C .

We will consider matrices in more detail later, but this will suffice for now. Let us get back to solving linear equations.

1.3 The Matrix Equation

Consider the system of equations in (1.1). We can write this as a matrix equation by using the laws of matrix multiplication:

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1c} \\ a_{21} & a_{22} & \dots & a_{2c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r1} & a_{r2} & \dots & a_{rc} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_c \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_r \end{pmatrix} \quad (1.6)$$

We abbreviate this as

$$A\mathbf{x} = \mathbf{b} \quad (1.7)$$

where A is the matrix of coefficients, $\mathbf{x}$ is the column vector of our variables:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_c \end{pmatrix}$$

and $\mathbf{b}$ is the column vector of constants:

$$\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_r \end{pmatrix}$$

Let us revisit [Example 1.1](#). The insight here is that systematically, if we have r -many equations, replacing the i -th equation with a linear combination of the r -equations does not change the system of the equation, and neither does swapping rows.

We say two systems of equations are *equivalent* if they've the same set of solutions.

Definition 1.2

To be a little more clear, the following operations retain the equivalence of system of equations:

1. Scale any of the equations by a scalar.

For now a vector is a member of $\mathbb{R}^n$ or $\mathbb{C}^n$. Later, we will give a more general definition, as an element of a vector space.

2. Replace eqn. i by eqn. $i + a$ eqn. j , for any $c \in \mathbb{F}$.
3. Swap eqn. i with row j .

Where 1 and 2 maybe condensed to just say replace an equation by a linear combination of all the equations, by first doing 1. and then 2.

Remark 1.3.1. Hoffman-Kunze defines equivalent systems to be where each equation in one system can be obtained as a linear combination of the equations in the second system. We can also say they're equivalent iff there exist a sequence of such operations that transforms one system into the other.

Now, we claim that such two systems must have the solutions. This follows rather straightforwardly, one can show that a solution of one system must be a solution of the other, and then the converse. This works because of linearity, and that equation is a linear combination of the equations of the other system'.

To understand how these operations transform the system of equations, we consider the augmented matrix $(A \mid b)$:

For the matrix equation $Ax = b$. We define the *augmented* matrix $(A \mid b)$:

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1c} & b_1 \\ a_{21} & a_{22} & \dots & a_{2c} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{r1} & a_{r2} & \dots & a_{rc} & b_r \end{array} \right)$$

Definition 1.3

Example 1.2

For the previous set of equations, the augmented matrix is:

$$\left(\begin{array}{cccc|c} 2 & 4 & 6 & 8 & 1 \\ 3 & 0 & 7 & 1 & 10 \end{array} \right)$$

The idea behind this is that the names of the variable is auxiliary, and this allows us to easily represent the operations which preserve the equivalence of the system of the equations.

The associated elementary row operations on the augmented matrix are:

1. Linear combination of the n rows.
2. Swapping the i and j rows.

Remark 1.3.2. Note that we can reduce a matrix to something of the form

We'll formally prove this later, but think about how you could do this in an algorithmic manner, it isn't that hard. But then why would we want to change our matrix to such a thing?

Consider the last row. If we converted our augmented matrix to the above form, it really just represents a system of equations. If we then consider the matrix equation, $A'\mathbf{x} = \mathbf{b}'$ (where A' and $\mathbf{b}'$ are obtained from the equivalent augmented matrix which is reduced to the above form from the original). Look at the last row, after doing the proper matrix-vector multiplication, we see that the last variable is fully determined. Of course, if there were some cells left after the 1, say the i -th cell is the 1 and it goes up till n , what would happen is that some variable, x_i would be determined in terms of $x_{i+1}, \dots, x_n$ (and a constant term). Then, we could substitute x_i to the equation obtained from the second last row, and determine x_{i-1} in terms of the same variables! Thus this form allows us to easily solve our system of equations and find the solution space.

The leading non-zero row in any row of the augmented matrix is called a *pivot*.

Definition 1.4

Intuitively this form means something like this—A matrix is said to be in a *row-echelon* form if “pivots in successive rows move to the right” for all rows except the last. Typically we scale to make the pivots 1. This is hardly a precise definition of course, so let us formally state it:

A $r \times c$ matrix A is in REF if the following is true for all $2 \leq i \leq r$. If (row i has a pivot), then (so must row $i - 1$ and left most non-zero entry in row $i - 1$ must be in an earlier column than “that” in row i).

Definition 1.5

We can formalise our intuition in Remark 1.3.2 as the following theorem:

Any matrix can be converted to row-echelon form through elementary row operations; equivalently there exists an *equivalent* REF matrix for any matrix.

Theorem 1.1

Algorithmically we can think about it like this—if we start with a zero-matrix we are vacuously done, but otherwise consider the *first* non-zero column. Swap the row containing that to the first row. Now operation (2), i -th row $+a \times j$ -th row helps us kill all non-zero entries in that column. Now we induct, the point is that we are done with the first row of the echelon form and can start looking for the second row by considering the *next* non-zero column in a smaller matrix (the red one), and converting it to a REF form won't impact the work we've already done.

Now, let us formally prove this.

$$\begin{pmatrix} 0 & \cdots & 0 & * & * & \cdots \\ \vdots & & \vdots & & & \\ \vdots & & \vdots & & & \\ 0 & \cdots & 0 & \boxed{} \end{pmatrix}$$

Proof. We induct on the number of the rows. For a one-row matrix, if all the columns are zero then the statement vacuously holds. Otherwise, let the pivot be $c \neq 0$. Scaling by $1/c$ reduces it to the desired form.

Suppose we have shown this for matrices of number of rows $1, 2, \dots, k$. Consider a matrix with $k + 1$ rows. Find the first column with a non-zero element (if this does not exist, the matrix is just the zero-matrix, so our statement is vacuously true). Swap the row containing this to the top. Now, scale it by $1/c$ where c is the value of the pivot. Consider now the rows below the first row. Suppose they've entries a_i is the pivot column. Then, subtract a_i times the first row from the i -th row, so that all of the entries in the pivot column are now zero except the first row. Note that, now we can just reduce a matrix of a smaller size (the submatrix formed by rows 2 through $k + 1$) of k rows to a row-echelon form.

Since we do this using elementary row operations, which only involve linear combinations and swapping, the columns containing 0 can not be changed, as swapping only changes 0 to another 0, and a linear combination of 0 entries is also 0. Therefore, reducing it to a row-echelon form makes the whole matrix in a row-echelon form now (as the next pivot is at least one column after the first pivot, because all columns in the submatrix have zero entries up until the pivot column, so the first pivot can only occur later (or never)). $\square$

Remark 1.3.3. One may consider augmented matrices ad-hoc representations for now, but there is a somewhat nice reason for considering them. These elementary row operations correspond to multiplication by a specific kind of invertible matrix (called an elementary matrix) i.e

$$E\mathbf{A}\mathbf{x} = E\mathbf{b} \iff \mathbf{A}\mathbf{x} = \mathbf{b}.$$

We will discuss them later, but in particular a property that our augmented matrix exhibits is:

$$E(\mathbf{A} \mid \mathbf{b}) = (E\mathbf{A} \mid E\mathbf{b}).$$

We can now look at [Example 1.1](#) in this manner. Let us show how the row operations correspond to matrix multiplication.

$$E = E_4 E_3 E_2 E_1 = \begin{pmatrix} 0 & 1/3 \\ 1/4 & -1/6 \end{pmatrix}, \quad \det E = -\frac{1}{12} \neq 0$$

$$\mathbf{A} = \begin{pmatrix} 2 & 4 & 6 & 8 \\ 3 & 0 & 7 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 10 \end{pmatrix}$$

$$E\mathbf{A} = \begin{pmatrix} 0 & 1/3 \\ 1/4 & -1/6 \end{pmatrix} \begin{pmatrix} 2 & 4 & 6 & 8 \\ 3 & 0 & 7 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 7/3 & 1/3 \\ 0 & 1 & 1/3 & 11/6 \end{pmatrix}$$

$$E\mathbf{b} = \begin{pmatrix} 0 & 1/3 \\ 1/4 & -1/6 \end{pmatrix} \begin{pmatrix} 1 \\ 10 \end{pmatrix} = \begin{pmatrix} 10/3 \\ -17/12 \end{pmatrix}$$

$$E(A \mid \mathbf{b}) = \begin{pmatrix} 0 & 1/3 \\ 1/4 & -1/6 \end{pmatrix} \begin{pmatrix} 2 & 4 & 6 & 8 & \mid & 1 \\ 3 & 0 & 7 & 1 & \mid & 10 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 7/3 & 1/3 & \mid & 10/3 \\ 0 & 1 & 1/3 & 11/6 & \mid & -17/12 \end{pmatrix}$$

Thus may think of augmented matrices as *representations*, or really, as the system of equations themselves, and then the way of writing it down as the $r \times (c + 1)$ table of numbers as a *representation*. One understands this as:

If the system is inconsistent, will the row-echelon form of the augmented matrix have rows of the form:

$$(\ 0 \ 0 \ \cdots \ 0 \mid c \)$$

where $c \neq 0$.

Yes, otherwise since we can reduce any augmented matrix to a row-echelon form, and this swapping back of variables will lead us to a set of solutions.

Question